Design Pattern Analysis in a Next.js-Based Video Streaming Application

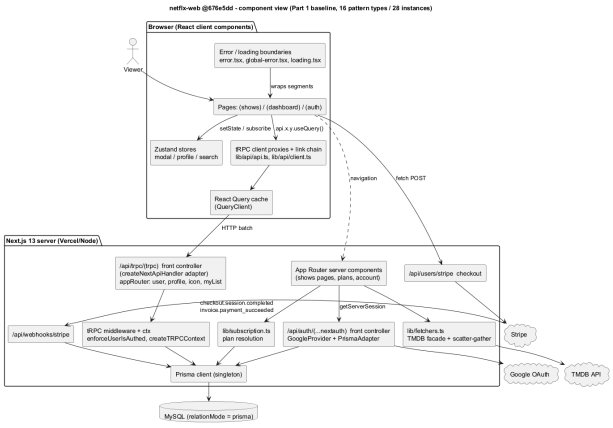
23CSE455 Design Patterns — Case Study, Team A12

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Application under study

- **netflx-web** (sadmann7), Netflix clone
- Next.js 13.4 app router, tRPC 10, NextAuth 4, Prisma 4 (MySQL), Stripe, Zustand, React Query, TMDB API
- Baseline commit 676e5dd (2024-06-12)
- No LICENSE file — academic analysis only
- Seven areas: user management, catalogue, subscription, recommendation, authentication, content access, caching



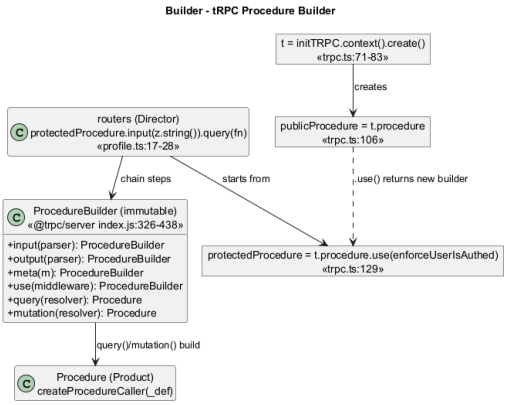
Part 1 — pattern catalogue at the baseline (two passes)

Pattern	Instances	#	Concr.
Chain of Responsibility	Authenticated Procedure Pipeline; **tRPC Client Link Chain**	2	3
Singleton	Shared Prisma Client; Stripe API Client	2	2
Observer	Show Modal Selection; Active Profile Selection; Search Results; **Search Input DOM Listeners**	4	17
Decorator	Persisted Profile Store (devtools(persist(...)))	1	3
Adapter	NextAuth Prisma Persistence Adapter; **tRPC Next.js API Handler Adapter**	2	2
Strategy	Sign-in Provider Strategy; **Wire Serialization Transformer**	2	2
Facade / Composite	TMDB Catalogue Facade / API Router Tree	1+1	1+5
Proxy	2 tRPC client proxies; Environment Secret Guard	3	3
Read-Through Cache	My List Query Cache; **Profile and Account Query Cache**	2	2
Front Controller	tRPC API Front Controller; NextAuth Auth Route Front Controller	2	2
Builder / Context Object	tRPC Procedure Builder / tRPC Request Context	1+1	2+1
Error / Suspense Boundary	Shows Segment + Root Global Error Boundary / Shows Loading Placeholder	2+1	2+1
Scatter-Gather	TMDB Row Scatter-Gather	1	2
16 types	first pass 10 types / 15 instances; second pass (bold) +6 / +13	**28**	**50**

11 rejected (e.g. dead caching proxy, Suspense around non-suspending forms), 5 uncertain (e.g. Stripe webhook). 156 cited ranges checked by script: 0 problems.

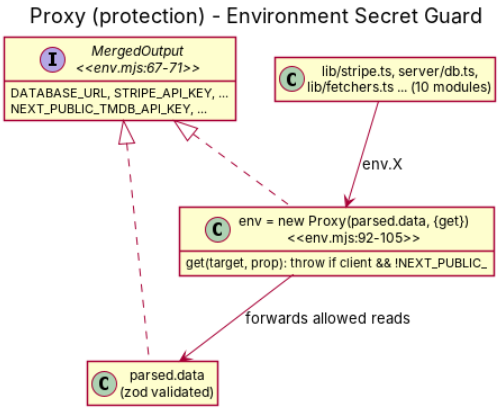
Second pass — what the area-driven first pass missed

- Read *every* in-scope file, not only area entry files
- Verified dependency roles in the exact versions (`npm pack @trpc/*` 10.28.0, next-auth 4.22.1, next 13.4.4)
- Found the infrastructure patterns: two **front controllers**, an immutable **builder**, a **context object**, error and **Suspense boundaries**, **scatter-gather**
- Corrected one existing record: the pages-router client's own link chain is never executed



Example: Environment Secret Guard (protection proxy)

- `env = new Proxy(parsed.data, {get})` in `env.mjs:92-105`
- On the client, any non-`NEXT_PUBLIC_` read throws
- Explains a smell found in Part 2: plan data was duplicated because the server list reads secrets



Part 2 — three changes

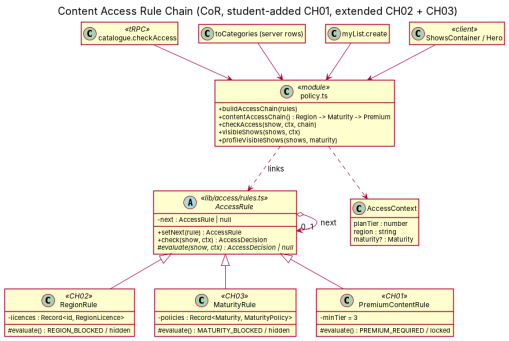
Change	What	Tests
CH01 premium + Ultra plan	tier-based premium titles, Ultra (7 profiles), subscription middleware	29/29
CH02 regional restrictions	licences per country, geo header, server-side row filtering	45/45
CH03 parental control	KIDS / TEEN / ADULT rating per profile	65/65

- Characterization tests written first and run on the *original* code (19/19)
- New student-added **Content Access Rule Chain**: Region → Maturity → Premium (hide before lock)
- Existing tRPC middleware chain extended with `attachSubscription`; immutable builders kept `protectedProcedure` unchanged

Pattern evolution

Snapshot	Types	Inst.	Concr.
baseline	16	28	50
CH01	17	30	56
CH02	17	30	57
CH03	17	30	60

- New: Content Access Rule Chain (CoR), Access Rule Evaluation Template (Template Method)
- Extended: auth pipeline, router tree, profile observers, tRPC front controller, procedure builder, request context
- Modified: TMDB facade, TMDB scatter-gather, search observer; none removed



Refactoring — real smells, measured with the same tool

Smell	Technique	Before	After
Duplicated plan data (UI vs server, already drifted)	shared client-safe catalogue	139 NLOC	123 (+1 plan)
Copy-pasted TMDB fetch blocks (two scatter-gathers)	rows as data + `fetchRows`	max CCN 9	3
Same row array in 3 pages	`GENRE_ROWS` + `toCategories`	174 NLOC	116
Duplicated zod schema / lookups in My List	Extract Variable / Function	169 NLOC	114

Whole `src/`: 6694 → 6916 NLOC (+3.3%), avg CCN 2.31 → 2.26. LOC alone does not prove better design; the evidence is behaviour preserved by tests.

Part 3 — four languages, one shared fixture, 16 pattern types

	Java	Python	JavaScript	C++
cases passed (16 problems)	177/177	177/177	177/177	177/177
total source LOC	899	568	623	1019
max function CC	11	9	10	11
median of median process times (ms)	116.8	50.1	59.2	16.2

- Proxy: JS `Proxy` trap ≈ source; Python `__getattr__`; Java `reflect.Proxy`; C++ plain class
- Builder: immutability by spread/`freeze` (JS), tuples (Py), `List.copyOf`/records (Java), value copies + const methods (C++)
- Scatter-Gather: `Promise.all` *is* the pattern in JS (10 NLOC); C++ starts real threads (`std::async`)
- Times (Windows 11) are start-up dominated: they compare runtimes, not patterns

LLM use and verification

- Session 1 (claude.ai, Claude Opus 5.5, 2026-10-10): Parts 1–3. Session 2 (Claude Code, Claude Opus 5.5, 2026-10-29/30): exhaustive second pass, 6 new Part 3 problems, full re-run
- Prompts: `Part1/Part1_prompt.txt`, `Part2/Part2_prompt.text`, `Part3/llm/part3_prompt.txt`; records in each part's `llm.csv`
- Errors found by checking: missed instances (env proxy; 13 infrastructure instances), wrong line ranges, a wrong transport role, an inactive link chain claimed active, three first code drafts that did not compile or follow React's reset rule
- All generated code is covered by tests (65 app tests, 708 cross-language checks)

Findings

- 16 pattern types / 28 instances at the baseline; most are inherited from frameworks (tRPC, zustand, NextAuth, React Query, Next.js) and configured, not written
- An area-driven first pass misses the infrastructure layer — read every file
- Recommendation has no pattern because it has no logic
- A chain of rules absorbed premium, region and parental control with one class each
- Pre-existing issues: client-side PIN check, missing ownership checks on profile / My List, dead caching proxy, two parallel tRPC clients
- Limits: no end-to-end run with real Stripe / TMDB / Google; sample licence data